



MECH MAINT- BOILER Department
SIMHADRI



Ref:SMTTP/MECH MAINT-/2022-23/BMDMILLS/779479

Date:

21/01/2023(10:28:42)

Subject:Approval for Revision of LMI on Guidelines for Coal Mill and Bunker Maintenance at NTPC Simhadri

- Location Management Instructions (LMI) for **"Guidelines for Coal Mill and Bunker Maintenance"** (attached) at NTPC- Simhadri, has been revised vide "LMI/OGN/OPS/MECH/052/ ISSUE NO.1; REVISION NO. 1.00, DATE: JAN'2023". As per LMI, 2-yearly review was due in December'22.
- The following modifications have been made in the doc:
 1. Name of the LMI has been modified to **"Guidelines for Coal Mill and Bunker Maintenance"**
 2. Clause 6.4 PERFORMANCE CHECKS, CHECKLIST –III, point number-4, frequency has been modified to 3 months from earlier six months.
 3. Clause 6.4 PERFORMANCE CHECKS, CHECKLIST –III, point number-5 has been added
 4. Clause 6.4 PERFORMANCE CHECKS, CHECKLIST –III, point number-10 has been added
 5. Clause 6.5 MILL HEALTH MATRIX CHECKLIST –IV has been added.
 6. Clauses 6.6 BEST PRACTICES IN MILLING SYSTEM - point no O to Y has been added.
 7. A new Clause No-7.0 "INSPECTION/ MAINTENANCE OF COAL BUNKERS" has been added.
 8. The LMI has been revised as per new document "OGN/OPS/MECH/052 Rev. No.: 1 Nov. 2022".
- Submitted for approval of BUH please.

Submitted by:

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On:23/01/2023(09:32:41) **Ref:**SMTPP/MECH MAINT-/2022-23/BMDMILLS/779479

The LMI has been revised. Necessary changes done in LMI are mentioned at page-1 of note.

Submitted for approval.

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On:23/01/2023(10:41:58) **Ref:**SMTPP/MECH MAINT-/2022-23/BMDMILLS/779479

May pl. see.

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Reviewed and found okay. Noted chemistry related points.

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On:28/01/2023(18:13:24) **Ref:**SMTTPP/MECH MAINT-/2022-23/BMDMILLS/779479

As peattached revised LMI, at page no. 10 unit of measurement may please be mentioned.

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On:29/01/2023(23:48:11) **Ref:**SMTTPP/MECH MAINT-/2022-23/BMDMILLS/779479

In attached revised LMI at page no 10, the unit of measurement is "mm" .

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On:06/02/2023(10:30:05) **Ref:**SMTTP/MECH MAINT-/2022-23/BMDMILLS/779479

Necessary changes regarding unit of measurement done at page no. 10 in LMI doc(attached).

Submitted for approval please

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Fpr review and further approval please.

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On:27/02/2023(12:05:27)

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TITLE: GUIDELINES FOR COAL MILL AND BUNKER MAINTENANCE		
REV: 1.00 DATE: JAN '23		
REF NO. LMI/OGN/OPS/MECH/052		

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/052/

ISSUE NO.1; REVISION NO. 1.00; DATE: JAN'2023

Guidelines for Coal Mill and Bunker Maintenance

APPROVED FOR

Implementation-----

HOP, SIMHADRI

Date-----

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LMI APPROVAL DOCUMENTS

<u>1. TITLE OF LMI</u>	Guidelines for Coal Mill and Bunker Maintenance
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/052/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV: 1.00; JAN'2023</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	COS-ISO-00-OGN/OPS/MECH/052 Rev. No.: 1 Nov. 2022
<u>4.PREPARED BY:</u> SR. MANAGER (BMD- Mills)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM (BMD-Mills)	SIGNATURE-----
AGM (BMD)	SIGNATURE-----
AGM (CHEM)	SIGNATURE-----
AGM (EEMG)	SIGNATURE-----
GM (MTC)	SIGNATURE-----
GM (O)	SIGNATURE-----
GM (O & M)	SIGNATURE-----
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE: ----- HOP, SIMHADRI
<u>7. DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.

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Guidelines for Coal Mill and Bunker Maintenance

2.0 INTRODUCTION

The document provides general guidelines on various checks to be carried out and correction / Rectifications to be taken up in XRP-1003 Mills and associated equipments viz. LOPs, Hoists and safety protection systems.

The results of a thorough inspection of Coal mill will enable the plant to properly plan the necessary repairs required and take up the maintenance activities. A properly maintained Coal mill will provide efficient, reliable and trouble-free operation.

3.0 SUPERSEDED DOCUMENTS

OGN/OPS/MECH/052 Rev. No.: 0 August 2020

4.0 SAFETY ASPECTS

- a.** All personnel that operate, inspect, or maintain Mills should be trained in the plant safety procedures.

- Responsibility : BMD

- b.** The Mill motor and mill auxiliaries must be electrical isolated & locked. The respective HAG, CAG, HAD, CAD, MDV, BIG's discharge damper/ gates, Seal air valve, inerting valves and Feeder outlet gates should be in CLOSED CONDITION & ISOLATED i.e. Mill should be isolated from air inlet side, raw coal entry side, Pulverized Fuel discharge side and mill motor side.

- Responsibility : Operation

- c.** Hot mill will distill combustible gases, from residual coal. Therefore, care to be taken when opening the mill doors. High VM Coal / Imported Coal may distill more combustible gases. Hence mill doors should only be opened when the temperature of Mill is less than 50 degree Celsius.

- Responsibility : BMD

- d.** Take care when opening mill inspection doors. The mill may be pressurized. Check that the hot and cold air shut off gates, the mill and feeder seal air valves and the pulveriser discharge valves are closed. The mill motor, breaker should be racked out and tagged. The pyrite hopper door should be carefully cracked and then completely opened. Do not open and remove the bolts/ nuts of door/ Manhole at once. Carefully remove the bolts of the inspection door partially & rap the flange of the door to break the seal and to equalize the pressure of

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and outside. Then remove the bolts and the door. Eye protection (glass or face shield) is recommended.

- Responsibility: BMD

- e.** All doors of Mills should be opened before entering inside the Mill.

-Responsibility: BMD

- f.** During Roller Setting (Ring Roll Adjustment by running the Mill) / Trial run no man shall be inside the Mill and shall be away from the Manhole door.

-Responsibility: BMD

- g.** Do not use 230 Volts supply lights inside the Mill, only 24 Volts supply to be used inside Mill.

-Responsibility: BMD

- h.** Mills are confined spaces that may be filled with harmful gases. Confined space entry procedures should be followed before entry.

-Responsibility: BMD/Chemistry

- i.** Before cancellation of PTW, it should be ensured that all personnel and tools are removed from the Mill.

-Responsibility: BMD

- j.** All coupling guards should be fitted in position before starting the Mill.

-Responsibility: BMD

- k.** Do not weld on the bowl or any part attached to it unless the welding machine is grounded to the bowl to prevent arching across bearing assemblies.

-Responsibility: BMD

- l.** Mandatory use of safety belts by people working at heights.

-Responsibility: BMD

- m.** Hoist in testing working condition, tested and within validity period before usage in overhauls.

-Responsibility: BMD

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5.0 RECORD OF MAINTENANCE

5.1 Maintenance record should contain major jobs carried out, bearing replacement details, equipment overhauling details, modification and deviation away from OEM recommendation if any.

5.2 Maintenance record should also be maintained in hard/soft copy.

5.3 Preventive Maintenance Checklist as provided in Annex 1 to be maintained for each PM.

6.0 INSPECTION / MAINTENANCE OF COAL MILL

6.1 CHECKS BEFORE START OF MAINTENANCE WORKS: CHECKLIST -I

6.2 MILL CLEARANCES

6.3 MAINTENANCE CHECKS/ SCHEDULE DURING MILL OH-CHECK LIST –II

6.4 PERFORMANCE CHECKS, CHECKLIST –III

6.5 MILL HEALTH MATRIX CHECKLIST –IV

6.6 BEST PRACTICES IN MILLING SYSTEM:

- a.** LMI/OD/OPS/SYST/009, issue no 1, Rev 3.0: The Safe operation and Maintenance of Pulverized Fuel Plant should be strictly followed to avoid Mill fire and ensuring Mill component healthiness.
- b.** In case of PF pipe-choking LMI/OGN/OPS/MECH-045, Issue No. 1, Rev 0 : Chocking of PF Pipes should be followed for clearing of PF pipes.
- c.** Mill Inerting System at all stations should be healthy and should remain in charged condition.
- d.** PF Pipe temperature measurement, wherever provided, should be restored. Wherever there is no provision, the same should be provided.
- e.** Mill rejects should be minimized by Timely replacement of Grinding elements, Maintaining adequate spring compression and spring gap, Adjustments based on Fineness checked by Isokinetic sampling of PF.
- f.** Air flow through Mill should be optimized and maintained as per air flow vs coal flow curve.
- g.** A committee consisting of members from BMD, Operation, EEMG and MTP (CMG) should be formed at each station to carry out root cause analysis of every failure and partial loading in rotary equipment (Mills).

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- h.** All outages in Mills (major breakdown) irrespective of whether outages result in partial loss or not, should be reported in a standard format (C2 Form). This will enable:
 - a) Understanding the damaged mechanism
 - b) Establishment of root cause
 - c) Formulation of action plan to prevent similar failures in future
 - d) Dissemination of learning among all stations of NTPC
- i.** If TA/ Expert visits the site for a particular job then a TA visit report consisting dismantling / assembling procedure, methodology & critical clearances etc., shall be prepared & to be shared with Head CC-OS Boiler. This report shall be circulated among other NTPC stations for knowledge sharing.
- j.** Load testing of lifting Tools-tackles, Mills Hoists, supporting structures by safety/ third party before usages to be done.
- k.** More emphasis should be given on on-line condition monitoring. Maintenance engineers and Shift charge engineer in every station must be equipped with a portable vibration analyzer and digital temperature recorder.
- l.** Before top up or filling new oil, moisture and MI may be checked.
- m.** When the mill is not in operation for more than a week, run the pulveriser in no load condition for 5 minutes, once a week. Lub oil should be started at least one hour before Mill start.
- n.** During monsoon / rainy season following to be ensured:
 - a) Mill outlet temperature to be maintained. In case of wet coal feed may be reduced otherwise too low Mill outlet temperature may choke the PF coal pipe.
 - b) Ensure no passing in CAD.
 - c) Healthiness of Insulation of PAPH and hot air ducts to be ensured to avoid ingress of water inside.
- o.** In case of Mill Motor coupling failure, alignment values as found after failure (i.e. before start of work) to be taken and record of same should be kept.
- p.** In case of Mill Motor coupling alignment, Motor shaft axial play also to be checked in addition to gear box and foundation.
- q.** As per EPRI after 90000 hrs (running hrs) Girth gear to be reversed and new pinion to be used. Hence after life of 90000 hrs backlash, root gap to be measured and blue matching to be done in each OH and record to be kept. In case any abnormality found the GG reversal & Pinion replacement to be planned.
- r.** In each Overhauling Inspection of all hoses to be carried for cracks, abrasions, flexibility

(Should not hardened) and localised swelling. Healthiness to be ensured by running LOP's and inspection for any leakage points after taking LOP out of service and proper permit.

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The length & design pressure rating of hoses must be as per OEM specification. All Mill hoses are to be mandatorily replaced in every 10 years.

- s. In case of high PQ/ moisture of Mill lub oil, Internal inspection to be done to find the root cause and Oil to centrifuge/replaced immediately.
- t. In each PM of Mill, complete opening and closing of MDV's to be checked.
- u. Midlife servicing of RJ assembly (i.e. Inspection of Brg, repl of Lub oil & Oil seals, checking of Float) may be planned based on average life of seals where Grinding Roll/Tyre life is very high compared to seal life.
- v. Labyrinth seal near PGB to be inspected and clearance to be checked every midlife serv and GR/BRS repl
- w. Upper radial bearing clearance to be checked with vertical shaft repl/ mechanical face seal repl and record to be maintained
- x. Mill health matrix as per annexure to be maintained and to be shared with COS every six month.
- y. PGB Gearbox servicing as per OEM recommendation

7.0 INSPECTION/ MAINTENANCE OF COAL BUNKERS:

During Unit OH Bunker inspection is to be carried out. Rolling plan for Inspection of Coal Bunkers to be made such that half of the bunker population should be inspected in each unit OH and during two consecutives OH all bunkers should be covered.

Following Circulars on bunker inspection are to be followed:

- Guidelines for Checking of Coal Bunker Hoppers (Conical Portion) made of SS-409M Ref No OS/BoP/00/Bunkers/ dated 16.05.2013
- Health assessment of Coal Bunkers (hoppers/conical portion + cylindrical portion) Ref No PEC/MCM/68/1 dated 02.02.2017

Inspection as below to be carried out after taking proper permit:

- I. Internal inspection and checking of the bunker after emptying the bunkers to be carried out.
- II. Inspection of shell hopper wall thickness or liner thickness
- III. Inspection and checking of all the joints from outside/ inside.
- IV. The checking of weld joints will include Visual Inspection and Die Penetration Test in the
 - (a) Dissimilar metal (MS-SS) joints, particularly ring beam joints,
 - (b) Hopper SS-SS joints.
- V. Site FQA may be involved for the checking and suitable protocol to be maintained. In case any major deviation is COS Boiler to be contacted.
- VI. Permanent Suitable platform / ladder to be made for ease of regular inspection in the ring beam joints from outside.
- VII. Based on visual inspection and test reports, an overall assessment to be done by the inspection team (BMD/FQA/FES) to derive whether hoppers require repairs/

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strengthening or replacement. For cases where damages are limited and local in nature, bunkers may be repaired. In case the damages observed are significant and widespread, team may recommend replacement of bunkers or a part or liner replacement in a phased manner and site may act accordingly.

VIII. Bunker Grill inspection to be done in case of foreign material

IX. In case of RSD –feeder to be emptied

8.0 REVIEW: The Head of O &M will be responsible for reviewing this document on a 2-yearly basis or as necessary.

9.0 DISTRIBUTION: The approved soft copy will be maintained in EEMG website as controlled copy

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10.0 MILL CLEARANCES: XRP 1003:

Sl No	Location (Description)	As per OEM (XRP-1003)^(mm)
01	Vertical gap and side gap between body liner segment and rotating vane segment of airport/louvre ring (fixed vanes) assembly.	8 to 11
02	Gap between inverted cone and inner cone	90- 100
03	Gap between center feed pipe lower and inner cone	100-114
04	Radial gap between airport ring deflector & mill side liner	30 (minimum)
05	Gap between Grinding Roll & BRS	6 to 8
06	Vertical gap between scraper and insulation cover plate	14 to 19
07	Radial gap between scraper and mill side bottom liner	25 to 35
08	Classifier blades/vanes position to be checked for uniform opening	120-160mm
09	Check the gap between spring assy. and journal head (to be uniform for all three rollers)/gap between Roller & grinding plate (grinding elements/Table), as applicable.	0.5mm

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CHECK LIST-I

11.0 Checks before Start of Maintenance Works:

Sl No	Check	Responsibility
01	Welding supply is available at site along with earthing strip near Mill.	EMD
02	Welding machine is Tested	EMD/BMD
03	24 V lighting supply available	BMD
04	Proper T&P and should be tested.	BMD
05	Hoist in working condition, tested and within validity dates (For major or OH jobs only)	BMD/MTP
06	Construction equipment like Truck, hydra, forklift for movement of material with valid LOAD TEST certificates	BMD
07	Proper approach, platform at locations where maintenance activity to be carried out.	BMD
08	Identification of leakage points/ patches in PF pipes/ MDV/MPO for replacement or maintenance work.	BMD/ OPERATION
09	Condition monitoring data available like vibration readings before any Major work	MTP

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CHECK LIST –II

12.0 Maintenance Checks/ Schedule during Mill Overhauling (Xrp-1003)

Sl No	Component	Check / Observation/ Description	Frequency
1.	Mill Drive Gear Housing/Thrust pad bearing level in case of Planetary GB	Check Oil level, check gauges, lines and headers for leakages,	Daily by OPN
2.	a. Drive coupling (BIBBY/ GEAR)	Inspection by opening removal and replacement of grease	Six monthly
	b. Lovejoy	Cleaning & Inspection	Each OH
3.	Motor alignment	Check Mill- Motor alignment - Planetary Gear Box	yearly / Any defect
		Worm-Wheel Worm Shaft gear box	Each Overhauling /Any defect
4.	Grinding Roll	Check all the rollers for free rotation	Every PM (Monthly)
5.	Grinding Roll Wear measurement	Check wear out with measurement for each roller	Monthly (or each PM)
6.	Grinding Roll	Replacement	OEM recommendation/ when Mill capacity reduced to less than 90 % of the design capacity for the design coal at rated fineness
7.	Roller Back inside mill	Clean all the pockets behind the roller near armour plates/ piece/ dam ring and remove foreign material if any	Every PM (Monthly)

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8.	RJ Spring Assembly	Check Spring compression	Each Overhauling / Any defect
9.	Scraper Chamber	Internal inspection and to check scrapers are intact	Every PM (Monthly)
10.	Air Seal Housing	Check oil seal, Clearance between vertical shaft and housing	Each Overhauling
11.	Roller Journal	Oil level	Monthly (or each PM)
12.	Lub Oil Filters	Check the oil filter of the oil supply system for clogging.	every 4 weeks
13.	Lube oil system (applicable for PGB only)	Check orifice choking in input shaft bearing	any defect
14.	Air filters / Breathers	Cleaning the air filters (breather filter) of both Lube oil system, of Gear box as applicable.	every 12 weeks
15.	Lub Oil Tank	Cleaning of lub oil tank if separate lub oil tank provided like in case of Planetary Gear Box	Condition based on NETRA report
16.	Mill Body	Arrest all the coal leakage points from the Mill body if any.	Every PM/any defect
17.	Worm Gear & Worm Shaft	Blue Matching of Worm Gear & Worm Shaft	Oil sampling report / noise / vibration
18.	Coal discharge Pipe	Coal pipe thickness measurement and its replacement to be done as per defects observed.	Each Overhauling
19.	fixed orifice	Ceramic items condition to be checked for any damages, for all orifices	Each Overhauling
20.	Body Liner Inspection	Inspection of all liner condition & thickness	Each PM
21.	Liner replacement	Replacement of all inside liner	Thickness/Condition based

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CHECK LIST III:

13.0 Performance Checks

Sl No	Check	Frequency
1.	Isokinetic sampling and Dirty air testing should be done in each Mill. Spring compression and classifier adjustment should be done in mills to achieve rated PF	Once in 6 Months / any defect like clinkering, etc.
	Fineness. Thereafter the existing sampling methods can continue. If required Mill compression can be increased and classifier can be further closed.	
2.	Unburnt carbon in Bottom Ash & Fly Ash	Monthly
3.	GCV of Mill reject	As per requirement
4.	Lub Oil Sample to be send to NETRA for oil analysis of each Mill/ WDA. In case of external lubrication oil sample for oil analysis has to be taken downstream the gearbox in the drain line of the lubrication system while the system is running. A suitable connection can be provided on the oil drain piping (for example an oil drain line with isolation valve and end plug). In case of internal lubrication sample to be taken from sampling point provided and not from bottom drain.	Once in 3 Months
5.	Moisture & MI of lube oil to be checked each Month	Monthly
6.	100 % Orifices in PF line inspection & replacement, if increase in Orifice Dia >1%	Each Unit Overhaul
7.	Clean air flow test should be conducted to verify PF line flow balancing	Each Overhaul
8.	Clearances as per attached List	Every PM
9.	Checks as per PM checks list	Every PM
10.	Mill capability test (up to PG test value)	Every Year

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CHECK LIST IV:

14.0 Mill Health Matrix

Mill	PQ Index date	PQ Index	Last Gear box / Worm gear or Worm shaft/ vertical shaft repl date	Details of repl of prev column	Detail of other Major brg repl done in Mill / Gear box	Date of work done of prev column	Date of last Bunker Inspection Including thickness & DP
A							
B							
C							
D							
E							
F							
G							
H							
J							
K							

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ANNEXURE-A

15.0 PM Checklist for Bowl type Coal Mill

PM Check List				
Equipment Name: Coal Mill			Date:	
S. N	Check	Allowable Limits / Recommendation	Observation	Status after PM
A.	Bowl Chamber			
1.	Roller oil level			
2.	Roller oil condition			
3.	Roller freeness			
4.	Roller oil leakage			
5.	Roller wear measurement			
6.	Journal head liners condition / replacement if required			
7.	Bowl hub cover & bolt condition			
8.	Check for Mill Stop Bolt Locking Plate is intact			
9.	Clamping ring & bolt condition			
10.	Classifier position setting			
11.	Classifier cone Condition Puncture			
12.	Classifier blades condition & opening			
13.	Inner cone (inside and outside) & its support pipes condition			
14.	Venturi Vane Condition			

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15.	Roller adjustment & gap between spring and journal head adjustment Roller-table gap checking & adjustment			
16.	Inverted Cone condition & Its gap			
17.	Center Feed pipe & Inner Cone Gap			
18.	Arrest all the coal leakage points from the Mill body if any			
B.	Scraper Chamber			
1.	Heat shield bolt condition			
2.	Lower /Upper skirt bolt Condition			
3.	Scraper condition			
4.	Scraper & insulation plate gap.			
5.	Bottom liner condition			
6.	Insulation cover plate condition			
7.	Air inlet vane condition (Hot vane)			
8.	Scraper man hole door condition.			
9.	Gear box oil level			
10.	Reject KGV Freeness & condition			
11.	Check the Mill reject path up to PV1 and clean/Remove the reject			
C.	Mill LOP			
1.	LOP pressure.			
2.	LOP seal & oil line leakage			
3.	Worm shaft gland leakage.			
4.	Cleaning of Mill gear box & Los breather			
5.	Clean both LO filter			
6.	Check the condition of all valves and fitting/ ferrule in LOP and arrest the leakage points if any			

